

Program Question_1

Implicit none

Real::a,b,p,fp,fa,fb,relative_error,error,p_prev=0.0

Integer::iteration=0

parameter(error=1.0E-8)

Open(unit=23,file='in2q1.txt')

Open(unit=27,file='out2q1.txt')

!Input the interval [a,b]

Read(23,*)a,b

!Compute f(a) and f(b)

fa=f(a)

fb=f(b)

if(fa*fb>0)then

Write(27,*)"There are no root in this interval"

end if

Write(27,*) "Iteration | a | b | p | f(p) | Relative Error"

!Bisection method

Do

!compute midpoint and functional value

p=(a+b)/2.0

fp=f(p)

!compute relative error

If(iteration>0)then

relative_error=abs((p-p_prev)/p)

else

relative_error=1.0

End If

!Update p_prev

p_prev=p

iteration=iteration+1

Write(27,'(I10,2F10.6,F10.6,F12.6,F12.6)')iteration,a,b,p,fp,relative_error

!termination

if(abs(fp)<error .Or. relative_error<error)Exit

!update interval

if(fa*fp<0)then

b=p

fb=fp

else

a=p

fa=fp

end if

end do

Write(27,*)"Root p=",p

!Function declaration

contains

Real function f(x)

Real,Intent(IN)::x

f = 3*x + SIN(x) - EXP(x)

end function

